IMPLEMENTING CONCEPT-BASED INTERPRETABILITY WITH TCAV

ImageNette target classification and DTD texture concepts

IML (AI60205) – Group Project Report

Group member	**Roll number**
Pareddi Sai Dinesh Reddy	24AI10041
Metla Gowtham Venkata Mahesh	24AI10037
Bokka Navadeep Pavan	24AI10019
Sabavath Arun Singh	24AI10005
Bhukya Saidulu	24AI10002

Department of Artificial Intelligence
Indian Institute of Technology Kharagpur

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1. Project Overview and Objectives

The objective of this project is to implement a concept-based interpretability method for a deep image classifier. A classifier can predict an object category accurately, but its internal decision process is not directly understandable from the output label alone. We use Testing with Concept Activation Vectors (TCAV) [1] to test whether human-describable visual concepts—in this project, textures—are reflected in the model's intermediate representations and influence its class scores.

We first fine-tune an ImageNet-pretrained ResNet-50 on the ten ImageNette classes. We then use examples from the Describable Textures Dataset (DTD) to define five texture concepts: dotted, striped, banded, zigzagged and woven. For each concept, a linear classifier learns a direction that separates concept activations from reference activations. This direction is used to calculate directional derivatives and TCAV scores for selected ImageNette classes.

The five predefined concept–class hypotheses are dotted→golf ball, striped→English springer, banded→parachute, zigzagged→chain saw and woven→cassette player. Experiments are performed at ResNet-50 layer3 and layer4, with layer3 designated as the primary analysis layer. Shuffled-label and random-reference controls are included to examine whether the measured sensitivity is specific to the intended concept.

What was implemented

- ImageNette data loading, stratified training/validation split and preprocessing.
- Fine-tuning and evaluation of a ten-class ResNet-50.
- DTD texture concept and reference-set preparation.
- Extraction of pooled layer3/layer4 activations.
- Logistic-regression CAV fitting and repeated runs.
- TCAV directional-derivative computation.
- Shuffled-label and random-reference control experiments.
- Statistical analysis, result tables and visualizations.

2. Datasets Used and Why They Were Selected

2.1 ImageNette – target classification dataset

ImageNette [4] is a ten-class subset of ImageNet. It was selected as the target dataset because it provides a manageable image-classification task with natural object images and established class labels. This makes it suitable for training a convolutional neural network and subsequently asking whether interpretable visual concepts affect its class scores.

The project uses the ImageNette training split for model development and the official validation split as the held-out test set. The training split is further divided into training and validation subsets using a stratified 90:10 split.

2.2 Describable Textures Dataset (DTD) – concept dataset

DTD [3] contains images organised around describable texture attributes. It was selected because TCAV requires examples of a human-understandable concept, and texture labels provide a direct way to construct such examples.

The five selected categories are dotted, striped, banded, zigzagged and woven. Each concept uses 120 images. These images are not used to train the ImageNette classifier; they are used to define concept directions in the classifier's activation space.

2.3 Reference pool

For each concept, a reference pool of 1,500 images is sampled from other DTD categories. Closely related categories listed in the configuration are excluded from the pool.

For true-concept CAV construction, the concept images form the positive group and the reference images form the negative group. The reference pool is also used in the random-reference control experiment, where two subsets of reference images are used to construct a CAV without using the selected concept images.

2.4 Image preprocessing

Images are converted to RGB, resized to 256 pixels and center-cropped to 224×224. They are then converted to tensors and normalized using ImageNet channel statistics.

The same deterministic resize, crop and normalization convention is used when preparing images for activation extraction. No additional training-time augmentation is claimed in this report.

3. Libraries and Their Uses

The implementation uses the Python deep-learning and scientific-computing ecosystem. Each library has a specific role in the pipeline.

Library	Purpose in this project
PyTorch (`torch`)	Tensor operations, model execution, automatic differentiation, activation extraction, batching, training loops, loss computation and checkpoint loading/saving.
`torchvision`	ResNet-50 architecture and ImageNet-pretrained weights; image datasets and image transformations.
NumPy	Numerical arrays and supporting numerical operations used in data handling and analysis.
Pandas	Loading, combining, filtering and exporting experiment results, CAV-run records and statistical result tables.
scikit-learn	Logistic-regression classifier for CAV fitting, train/validation splitting utilities and evaluation-related utilities.
SciPy	Statistical tests and numerical routines used in the analysis, including Mann–Whitney U and Welch-type comparisons.
statsmodels	Multiple-comparison correction, including Holm and Benjamini–Hochberg procedures.
Matplotlib	Generating training curves, confusion matrices, heatmaps and hypothesis-comparison plots.
Seaborn	Statistical plotting and presentation of distributions and heatmaps where used.
Pillow (PIL)	Opening, converting and handling image files during dataset preprocessing.
tqdm	Progress bars for long-running data and experiment stages.
Requests	Downloading the ImageNette and DTD dataset archives.
pytest	Running automated tests for the data pipeline, feature extraction, derivative calculations and related components.

4. Algorithms Implemented

The central part of the project is the implementation of Testing with Concept Activation Vectors (TCAV). The complete method has two stages: first, learn a direction representing a human-named concept in the network's activation space; then measure whether moving in that direction locally increases a chosen class score.

4.1 ResNet-50 fine-tuning

ResNet-50 [2] is a convolutional neural network with residual connections. We initialize it with ImageNet-pretrained weights and replace its final fully connected layer for ten ImageNette classes.

Images are passed through the network in mini-batches. Cross-entropy loss compares the predicted logits with the ground-truth labels, gradients are calculated by backpropagation, and the optimizer updates the trainable parameters using AdamW.

Validation accuracy and loss are monitored, and the best checkpoint is retained. This trained model is the function whose internal representations and class-score sensitivities are analysed by TCAV.

4.2 Activation extraction and representation

For each selected DTD concept image and reference image, the trained ResNet-50 is run in evaluation mode. Activations are captured at layer3 and layer4, and global average pooling is applied to obtain a fixed-length vector for each image.

The layer3 vector has 1,024 features and the layer4 vector has 2,048 features. Pooling gives a consistent vector representation that can be used by a conventional linear classifier to learn a concept direction.

4.3 Learning Concept Activation Vectors (CAVs)

For one concept C, its DTD activation vectors form the positive class (label 1), while activation vectors from the reference pool form the negative class (label 0).

A logistic-regression classifier is fitted to distinguish the two groups. The classifier's learned coefficient vector defines the separating direction in activation space. We normalize this vector and orient its sign so that the positive direction corresponds to increasing evidence for the named concept. This unit vector is the Concept Activation Vector, written v_C.

A validation split is used to check whether the concept is separable from the reference set. Runs below the configured 0.70 validation-accuracy threshold are not treated as reliable CAV fits.

Validation accuracy is the configured acceptance measure for CAV runs. The experiment repeats true-concept CAV fitting 30 times per concept and layer to measure variability from sampling and fitting.

4.4 TCAV directional derivative and score

Let x denote an input image, $h(x)$ its selected-layer representation, $f_k(h)$ the target model's score for class k, and v_C the CAV for concept C.

The directional derivative is

$$S_{C,k}(x) = \nabla_h f_k(h(x)) \cdot v_C.$$

It measures the local rate of change of the class score when the representation is moved a small amount in the concept direction. A positive derivative means that this small movement increases the class score locally.

The TCAV score is the fraction of evaluated examples with a positive derivative:

$$\mathrm{TCAV}(C,k) = \frac{|\{x \in X_k : S_{C,k}(x) > 0\}|}{|X_k|}.$$

A score close to 1 means the class score tends to increase along the concept direction for most evaluated examples; a score close to 0 means it rarely does.

This is a local sensitivity measurement, not proof that the model causally uses or explicitly recognizes the concept.

4.5 Layer-specific implementation

The project evaluates layer3 (1,024-dimensional pooled representation) and layer4 (2,048-dimensional pooled representation).

For layer3, the pooled CAV direction is mapped back to the spatial activation map as a spatially uniform shift: the channel-wise CAV direction is broadcast across the spatial positions. The directional derivative is evaluated with respect to the layer3 activation tensor using PyTorch automatic differentiation through the downstream network operations, so the gradient includes the subsequent layer4 processing and classifier.

The dot product of this activation gradient with the broadcast direction gives the directional response. This is an explicit spatially uniform intervention and should not be interpreted as pixel-level localization.

Layer4 is evaluated at the pooled representation immediately before the linear classifier head. Since the classifier head is linear, its gradient with respect to this representation is independent of the individual image for a given class. This can make layer4 TCAV scores degenerate at zero or one.

Layer3 is therefore used as the primary analysis layer.

4.6 Control algorithms

Shuffled-label control:

Concept/reference labels are shuffled before fitting the linear classifier. This tests whether apparent directional effects can arise without the intended concept-label relationship.

Random-reference control:

The random-reference control constructs CAVs using two randomly sampled subsets of the reference pool. One subset is assigned positive labels and the other negative labels.

Unlike the true-concept CAVs, this control does not use the selected concept images. It provides a baseline for measuring the influence of directions learned from arbitrary distinctions within the reference pool.

The true-concept TCAV scores are compared against the TCAV scores obtained from these random-reference CAVs.

These two controls test different aspects of the method: shuffled-label controls disrupt the label relationship, while random-reference controls evaluate directions learned from reference images alone. They should not be treated as interchangeable.

4.7 Experimental repetitions

The full experiment uses:

- 30 true-concept CAV runs per concept and layer.
- 50 shuffled-label control runs per concept and layer.
- 50 random-reference control runs per layer.

Repeated runs are used to evaluate the robustness of the learned concept directions and their resulting TCAV scores.

4.8 Statistical analysis

The primary confirmatory comparison is a two-sided Mann–Whitney U test between true-concept TCAV scores and the random-reference control distribution.

The resulting p-values quantify differences across the repeated CAV-fitting procedure under this particular sampling setup. Because runs can share evaluation images and overlapping reference samples, they should not be interpreted as evidence from independent datasets or as establishing population-level

generalization.

Welch's t-test and Mann–Whitney tests on mean directional derivatives are included as secondary analyses. Cliff's delta is used as an effect-size measure.

Holm correction controls family-wise error across the confirmatory comparisons, and Benjamini–Hochberg adjusted values are also reported for the relevant analysis families. The significance threshold is 0.05.

5. Implementation Challenges and How They Were Addressed

5.1 Combining datasets with different roles and distributions

ImageNette contains natural object photographs, whereas DTD contains texture-focused images. The two datasets therefore serve different purposes and have different visual distributions.

We kept ImageNette as the target-classification dataset and DTD as the source of concept and reference examples. A consistent image preprocessing pipeline was used for activation extraction. The remaining domain gap is treated as a limitation rather than assumed away.

5.2 Obtaining reliable concept directions

A CAV is useful only if its direction captures the intended concept rather than an unstable separation in the sampled activations.

We used logistic regression, normalized and consistently oriented its coefficient vector, evaluated CAV validation accuracy, and applied a configured minimum validation-accuracy threshold of 0.70.

Repeated true-concept CAV fitting (30 runs per concept and layer) was used to examine variability caused by sampling and fitting. These repeated runs assess fitting stability; they do not constitute independent replications on new test datasets.

5.3 Separating concept sensitivity from arbitrary directions

A high TCAV score alone is not enough to establish that a particular concept is responsible for a class-score response.

We implemented shuffled-label controls to break the concept/reference label relationship and random-reference controls to measure the sensitivity of directions learned from random subsets of the reference pool.

The confirmatory comparison is made against control distributions, not by interpreting a raw score in isolation. These controls address specific sources of misleading interpretation, although they cannot eliminate every possible confound.

5.4 Mapping pooled CAVs to spatial activations

CAVs are learned using globally pooled activation vectors, while layer3 produces a spatial feature map. The pooled direction therefore does not directly specify a unique spatial direction at every location.

In this implementation, the pooled CAV direction is broadcast uniformly across the layer3 spatial map for the directional-response calculation. This provides a defined intervention for the experiment, but it is not a pixel-level explanation or localization map.

5.5 Understanding near-binary layer4 scores

At layer4, the pooled representation is immediately followed by a linear classification head. The gradient of a linear class score with respect to its input representation is constant for a given class.

Consequently, the sign of the directional derivative can be the same across many or all evaluated images, producing TCAV scores close to zero or one. We retained layer4 as a comparison and treated layer3 as the primary analysis layer, where the preceding nonlinear feature processing provides a more informative comparison for this setup.

5.6 Statistical reliability and multiple comparisons

Testing several concept–class hypotheses creates a multiple-comparison problem.

We specified five hypotheses in advance, used a two-sided Mann–Whitney U test for the primary comparison, reported Cliff's delta as an effect size, and applied Holm correction to the confirmatory p-values.

Secondary tests and exploratory comparisons are interpreted separately. Repeated CAV runs share the same underlying evaluation examples, so the repeated-run count is not treated as an equivalent number of independent test datasets.

5.7 Computational cost and reproducibility

The full experiment involves repeated activation extraction, CAV fitting, gradient calculations and statistical comparisons. This made execution computationally expensive, particularly on CPU.

We organized the workflow into stages, used fixed configuration and random seeds, saved model checkpoints and structured result files, and ran the complete experiment in a Colab environment. These practices make the experiment easier to resume and inspect, although runtime can still vary with hardware and environment.

6. Limitations, Reproducibility and Practical Lessons

6.1 Limitations of the current study

Dataset domain gap.

The concept images come from DTD, while target-class evaluation images come from ImageNette. The concept direction may capture properties specific to the texture dataset or its image composition. A future study could test multiple concept sources or construct texture examples in the same photographic domain as the target images.

Finite concept coverage.

Each concept uses 120 images and a finite reference pool. Textures such as woven or striped can appear in many forms. The selected examples may not capture all valid appearances of the concept. More diverse examples and additional reference sampling would help test sensitivity to concept-set composition.

Repeated runs are not independent population replications.

Repeating CAV fitting estimates variability due to the fitting and sampling procedure. The same underlying test images are reused, so the run count must not be interpreted as an equivalent number of independent test datasets. Generalization to new datasets would require additional independently collected evaluation data.

TCAV is a local sensitivity measure, not a causal proof.

A positive directional derivative means that a small movement in a representation along the CAV direction locally increases the target class score. It does not prove that the network explicitly recognizes the concept, that the concept is necessary for the prediction, or that removing the concept from the input would change the prediction.

Layer and intervention dependence.

TCAV scores depend on the selected network layer and on how a pooled CAV direction is applied to the activation representation. Layer4's linear head leads to near-binary behaviour in this setup, while the layer3 calculation uses a spatially uniform intervention. Results should therefore be interpreted in the context of these implementation choices.

Limited hypothesis coverage.

Only five concept–class pairs were specified. The results do not establish that these are the only relevant textures for the ten ImageNette classes, nor do they provide an exhaustive explanation of the classifier.

6.2 Reproducibility practices

The implementation uses a fixed seed and saved configuration, checkpoints, training history, per-run CAV records, TCAV results, statistical summaries and figures.

The workflow is divided into data preparation, model training, activation/CAV computation, TCAV scoring and result analysis. This separation supports debugging and allows completed stages to be reused without repeating the entire pipeline.

6.3 Challenge-to-solution summary

Challenge	Implementation response
Different dataset distributions	Kept ImageNette for target classification and DTD for concept/reference examples; used consistent preprocessing.
CAV stability	Logistic regression, normalized/oriented directions, validation threshold and repeated fitting.
Arbitrary-direction effects	Shuffled-label and reference-versus-reference random-reference control experiments.
Pooled-to-spatial layer mapping	Explicitly broadcast the pooled layer3 direction uniformly over the spatial map.
Layer4 near-binary behaviour	Explained by the linear classifier head; layer3 retained as the primary analysis.
Multiple comparisons	Predefined hypotheses, two-sided Mann–Whitney U, effect sizes and Holm correction.
Expensive experiment	Fixed configuration, saved checkpoints/results and staged execution.

7. Implementation Evidence and Main Results

The trained target model achieved 99.79% validation accuracy and 99.24% test accuracy in the latest reported run.

The model was trained for 6 epochs using a learning rate of 0.0001, batch size 64, weight decay 0.0001 and random seed 42. The training configuration used ImageNet-pretrained ResNet-50 weights and a stratified 90:10 split of the ImageNette training data for training and validation.

The layer3 confirmatory results compare true-concept CAVs with the random-reference control distribution.

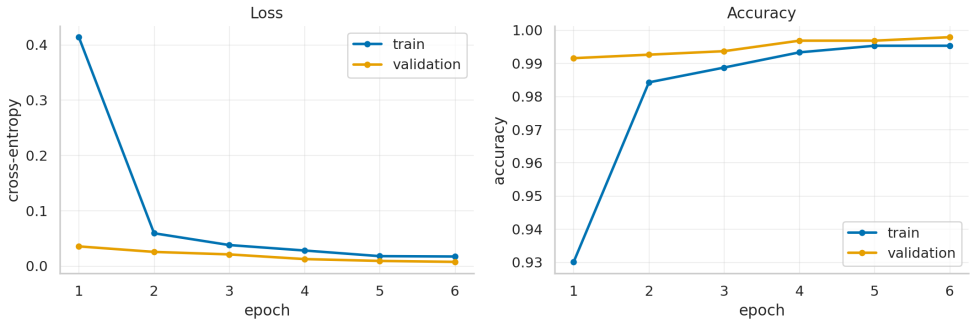


Figure 1: Training and validation curves for the fine-tuned ResNet-50.

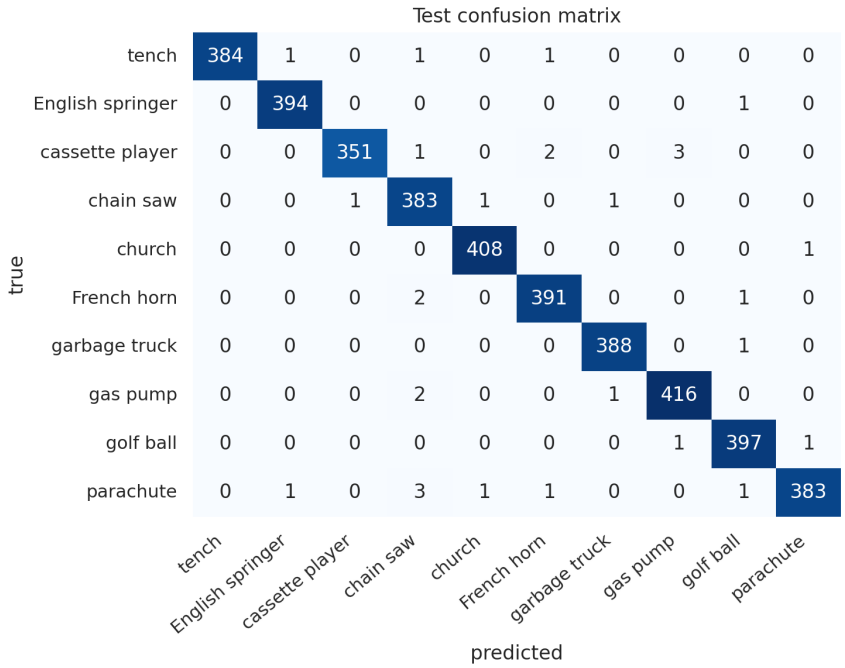


Figure 2: Test-set confusion matrix for the ten ImageNette classes.

Only dotted→golf ball and banded→parachute are significantly above the random-reference control after Holm correction. The other three hypotheses are not supported; their numerical values below control should not be interpreted as statistically established negative effects.

Hypothesis	**Mean true TCAV**	**Mean reference**	**Holm-adjusted p**	**Conclusion**
dotted → golf ball	0.889	0.537	8.11×10^{-12}	Supported
striped → English springer	0.392	0.462	0.650	Not supported
banded → parachute	0.822	0.476	4.99×10^{-12}	Supported
zigzagged → chain saw	0.413	0.501	0.158	Not supported
woven → cassette player	0.417	0.477	0.650	Not supported

Table 1: Mean layer3 TCAV scores across repeated runs and comparison with the random-reference control distribution. Values are rounded; Holm-adjusted p-values are from the confirmatory tests.

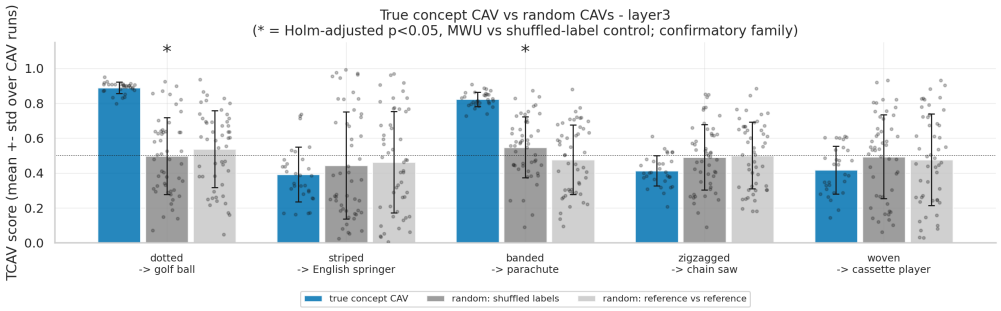


Figure 3: Layer3 comparison of the five hypothesized true concepts against random-reference controls.

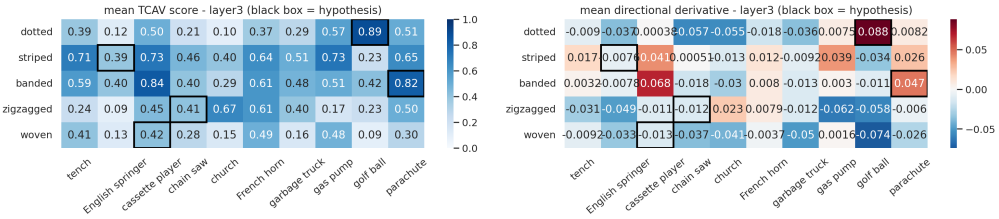


Figure 4: Layer3 heatmap showing concept sensitivity across target classes; included as supporting visual evidence.

8. Group Contributions, Conclusion and References

8.1 Individual roles and contributions

The group members contributed to the development, implementation, experimentation and documentation of the TCAV project. The responsibilities are summarized below.

Member	Contribution
Pareddi Sai Dinesh Reddy 24AI10041	Contributed to project coordination and integration, TCAV implementation, activation extraction, CAV construction, and analysis of experimental results.
Bokka Navadeep Pavan 24AI10019	Core algorithm implementation support, including the ResNet-50 feature and activation workflow, CAV fitting pipeline, repeated runs, and debugging of the concept-versus-reference classification stage.
Metla Gowtham Venkata Mahesh 24AI10037	Core algorithm implementation support, including the TCAV score and directional-derivative workflow, control experiments, statistical comparisons, and checking the layer-wise results.
Sabavath Arun Singh 24AI10005	Dataset preparation and preprocessing support, experiment execution checks, result organization, and validation of generated plots and tables.
Bhukya Saidulu 24AI10002	Documentation and presentation material, organizing implementation evidence, reviewing outputs, and assisting with final report preparation.

8.2 Conclusion

The project implemented a complete concept-based interpretability workflow using ImageNette, DTD, ResNet-50 and TCAV.

ImageNette provided the target classification task, while DTD supplied texture concepts and a reference pool. The implementation covered model fine-tuning, intermediate activation extraction, logistic-regression CAV learning, directional-derivative computation, control experiments and corrected statistical testing.

The primary layer3 analysis supported two of the five predefined hypotheses: dotted→golf ball and banded→parachute. The remaining hypotheses were not statistically supported.

The project demonstrates both the practical implementation of TCAV and the importance of carefully chosen controls and statistical interpretation.

8.3 References

1. Kim, B., et al. (2018). *Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV).* Proceedings of the 35th International Conference on Machine Learning (ICML).

2. He, K., Zhang, X., Ren, S., and Sun, J. (2016). *Deep Residual Learning for Image Recognition.* Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

3. Cimpoi, M., Maji, S., Kokkinos, I., Mohamed, S., and Vedaldi, A. (2014). *Describing Textures in the Wild.* Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

4. ImageNette Dataset. `https://github.com/fastai/imagenette`

5. PyTorch Documentation. `https://pytorch.org/`

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